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The high-frequency impact of news on long-term yields and forward rates: Is it real? ☆

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ABSTRACT

Macroeconomic news announcements move yields and forward rates on nominal and index-linked bonds and inflation compensation. This paper estimates the reactions using high-frequency data on nominal and index-linked bond yields, allowing the effects of news announcements on real rates and inflation compensation to be parsed far more precisely than is possible using daily data. Long-term nominal yields and forward rates are very sensitive to macroeconomic news announcements. Inflation compensation is sensitive to announcements about price indices and monetary policy. However, for news announcements about real economic activity, such as nonfarm payrolls, the vast majority of the sensitivity is concentrated in real rates. Accordingly, most of the sizeable impact of news about real economic activity on the nominal term structure of interest rates represents changes in expected future real short-term interest rates and/or real risk premia rather than changes in expected future inflation and/or inflation risk premia. Such sensitivity of real rates to macroeconomics news is hard to rationalize within the framework of existing macroeconomic models.

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1. Introduction

It has long been recognized that the high-frequency reaction of asset prices to macroeconomic news announcements represents a rich source of information about the effects of economic fundamentals on financial markets. Economic theory – and much empirical evidence – indicates that macroeconomic news announcements lead to sudden jumps in the conditional means of asset prices. Studying the relationship between these jumps and the surprise components of the announcements may be the closest thing that can be obtained in macroeconomics to a natural experiment. In this work, high-frequency data are especially valuable. In a short enough window around a news announcement, the data release should be the *only* information hitting the market. But macroeconomic news announcements explain only a modest share of the total variation in most asset prices, and so in longer windows the relationship between news announcements and asset prices will be clouded by the impact of other public or private information. For example, exchange rates are notoriously hard to relate to any macroeconomic fundamentals. Nonetheless, Andersen et al. (2003) find that in short

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intradaily windows around macroeconomic news announcements, there is a systematic relationship between macroeconomic surprises and exchange rate changes. This relationship cannot be detected at lower frequencies, because macroeconomic announcements explain only a small share of the total variability of exchange rates.

The literature on the reaction of financial markets to macroeconomic news is vast, and no attempt is made to survey it here. Many authors have studied the effects of news announcements on the term structure of interest rates, including Fleming and Remolona (1999), Gürkaynak et al. (2005a), and Goldberg and Klein (2005). A striking finding of these papers is that long-term nominal yields and forward rates are remarkably sensitive to incoming data. The release of an employment report can sometimes cause ten-year yields and forward rates to jump over 10 basis points. Nominal yields and forward rates embody inflation expectations, expected future real interest rates and term premia. Gürkaynak et al. (2005a) assume that news announcements do not affect long-term expectations of future real short-term interest rates or distant-horizon term premia. Hence, they interpret the sensitivity of far-horizon nominal forward rates to news announcements as investors revising their long-run inflation expectations in light of today's data, indicating that inflation expectations are very poorly anchored.

However, the sensitivity of nominal yields and forward rates to news surprises may owe to more than just inflation expectations: revisions to expected future real short-term interest rates or to nominal term premia (in turn the sum of inflation risk premia and real term premia) are also candidates, and there are no strong reasons to rule them out *a priori*. Beechey (2007) used a three-factor affine term structure model to decompose the variation in nominal forward rates in the wake of news announcements into nominal term premia and expectations of future nominal short-term interest rates, and argued that most of the reaction at long horizons was in nominal term premia.

This paper employs a decomposition of the reaction of nominal yields to news into real and inflation components. This decomposition is implemented using a new dataset of intradaily quotes on Treasury Inflation Protected Securities (TIPS) – debt securities for which the coupon and principal payments are indexed to the Consumer Price Index (CPI) – in combination with intradaily data on nominal yields. Intradaily changes in nominal yields, real yields, and the spread between these two – inflation compensation – are regressed on the surprise components of macroeconomic news announcements. Several recent papers, among them Gürkaynak et al. (2009a) and Beechey et al. (2007), have employed daily-frequency inflation compensation to estimate news-announcement regressions for the United States and other countries, highlighting the greater sensitivity of US forward rates of inflation compensation to various types of macroeconomic news announcements.

Macroeconomic announcements can be grouped into three broad types: news about prices, news about the real side of the economy (indicators of the labor market, spending, production and sentiment) and news about monetary policy. Armed with intradaily data, we find that different types of news have quite different effects on real rates and rates of inflation compensation. To summarize, price data mainly affect inflation compensation, real-side data have a large impact on real yields and forward rates – but little effect on forward rates of inflation compensation – and surprise tightenings of monetary policy push real rates up but inflation compensation down. Sizeable efficiency gains can be obtained from running these regressions with intradaily data, rather than data at the daily frequency. The fact that news buffets forward rates of inflation compensation is consistent with the conclusions of others that either long-run inflation expectations or inflation risk premia, or both, are sensitive to news—but the sensitivity appears to be confined to news about prices and monetary policy. Instead, our findings point to a great deal of sensitivity of the real component of long-run interest rates, a possibility that has largely been overlooked in the recent news-announcement literature. This suggests that the puzzling sensitivity of nominal forward rates lurks not just in the inflation component but, importantly, in uncertainty about the equilibrium real rate of interest. This in turn poses a substantial challenge to widely used dynamic stochastic general equilibrium models, such as those of Smets and Wouters (2007) and Christiano et al. (2005), which imply that the long-run expected value of the real short-term interest rate is constant.

The plan for the remainder of this paper is as follows. Section 2 introduces the data, Section 3 contains the main results and Section 4 offers some interpretation of the results and concluding thoughts. Some robustness checks and tests for structural stability are included in an online Appendix A.

2. The data

This section briefly describes data sources and construction. Fourteen macroeconomic announcements, including the surprise element of FOMC decisions about the target Federal Funds rate, are considered. Table 1 lists the announcements, their frequency and release times. For all announcements other than the target funds rate, the surprise component of each announcement is measured as the difference between the actual released value and the median prediction from the survey conducted by Money Market Services (MMS) on the previous Friday. For the target Fed Funds rate, the surprise component of the Fed's decision is measured using intradaily changes in Federal Funds futures prices, following the methodology of Kuttner (2001). Surprises are signed such that positive surprises represent stronger-than-expected growth or higher-than-expected inflation: for the unemployment rate and initial jobless claims, both of which are countercyclical indicators, the sign of the surprise is flipped so that positive surprises reflect stronger-than-expected growth for these as well. The surprises are then standardized by their respective standard deviations. Letting $A_{j,t}$ denote the released value of an

Table 1
US Macroeconomic announcements.

Data release	Source	Frequency	Release time	Standard deviation	Units
Capacity	Fed	Monthly	9:15	0.31	Percent
Confidence	Michigan	Monthly	10:00	4.84	Index
Core CPI	BLS	Monthly	8:30	0.09	Percent change mom
Durable goods	Census	Monthly	8:30	3.15	Percent change mom
ECI	BLS	Monthly	8:30	0.24	Percent change mom
GDP (advance)	BEA	Quarterly	8:30	0.81	Percent change qoq, ar
Claims	ETA	Weekly	8:30	18.5	Thousands
NAPM index	NAPM	Monthly	10:00	2.09	Index
Nonfarm	BLS	Monthly	8:30	92.5	Thousands
New homes	Census	Monthly	10:00	73.5	Thousands
Core PPI	BLS	Monthly	8:30	0.29	Percent change mom
Retail sales	Census	Monthly	8:30	0.70	Percent change mom
Unemployment	BLS	Monthly	8:30	0.13	Percent
FOMC	Fed	8 per year	14:15	5.3	Basis points

Notes: Acronyms for the sources are as follows: BEA (Bureau of Economic Analysis), BLS (Bureau of Labor Statistics), Census (Bureau of the Census), ETA (Employment and Training Administration), Fed (Federal Reserve Board of Governors), Michigan (University of Michigan), NAPM (National Association of Purchasing Managers). Acronyms for the units are: mom (month-over-month), qoq (quarter-over-quarter), ar (annualized rate). Standard deviations pertain to the sample February 2004–June 2008.

Table 2
TIPs bonds used in the analysis.

Date range	TIPS security used
Five year TIPS	
February 17, 2004–July 16, 2004	January 15, 2009, $3\frac{7}{8}\%$
Until October 26, 2004	January 15, 2010, $4\frac{1}{4}\%$
Until April 25, 2006	April 15, 2010, $\frac{7}{8}\%$
Until April 24, 2007	April 15, 2011, $2\frac{3}{8}\%$
Until April 22, 2008	April 15, 2012, 2%
Since April 23 2008	April 30, 2013, $\frac{5}{8}\%$
Ten year TIPS	
February 17, 2004–July 8, 2004	January 15, 2014, 2%
Until January 13, 2005	July 15, 2014, 2%
Until July 14, 2005	January 15, 2015, $1\frac{5}{8}\%$
Until January 12, 2006	July 15, 2015, $1\frac{7}{8}\%$
Until July 13, 2006	January 15, 2016, 2%
Until January 11, 2007	July 15, 2016, $2\frac{1}{2}\%$
Until July 12, 2007	January 15, 2017, $2\frac{3}{8}\%$
Until January 10, 2008	July 15, 2017, $2\frac{5}{8}\%$
Since January 11, 2008	July 15, 2018, $1\frac{5}{8}\%$

Notes: This table shows the specific TIPS with remaining maturities of approximately five and ten years in our intradaily dataset. The corresponding nominal securities are in all cases the most recently issued five- and ten-year notes. In most cases, the TIPS are also on-the-run securities, but the first two five-year TIPS are actually old ten-year TIPS with about five years left to maturity, because no five-year TIPS were issued by the Treasury between July 1997 and October 2004.

announcement of type j at time t, and $E_{j,t}$ denote the *ex-ante* expectation of this release, the surprise is defined as

$$s_{j,t} = \frac{A_{j,t} - E_{j,t}}{\sigma(A_{j,t} - E_{j,t})}$$

where $\sigma(\cdot)$ denotes the standard deviation. This transformation has the effect of making the units comparable across different types of announcements. Table 1 also lists the standard deviation of the surprises for each announcement type.

The high-frequency dataset consists of 5 min quotes of yields-to-maturity on nominal Treasury securities and TIPS with around five and ten years remaining time to maturity collected at the Federal Reserve Board. The specific bonds used are listed in Table 2 and are generally the most recently issued "on-the-run" five- and ten-year nominal Treasury notes and TIPS.[1] The data are available from February 17, 2004 to June 13, 2008. From these, the approximation of Shiller et al. (1983)

[1] An on-the-run security is the most recently issued bond of a given maturity.

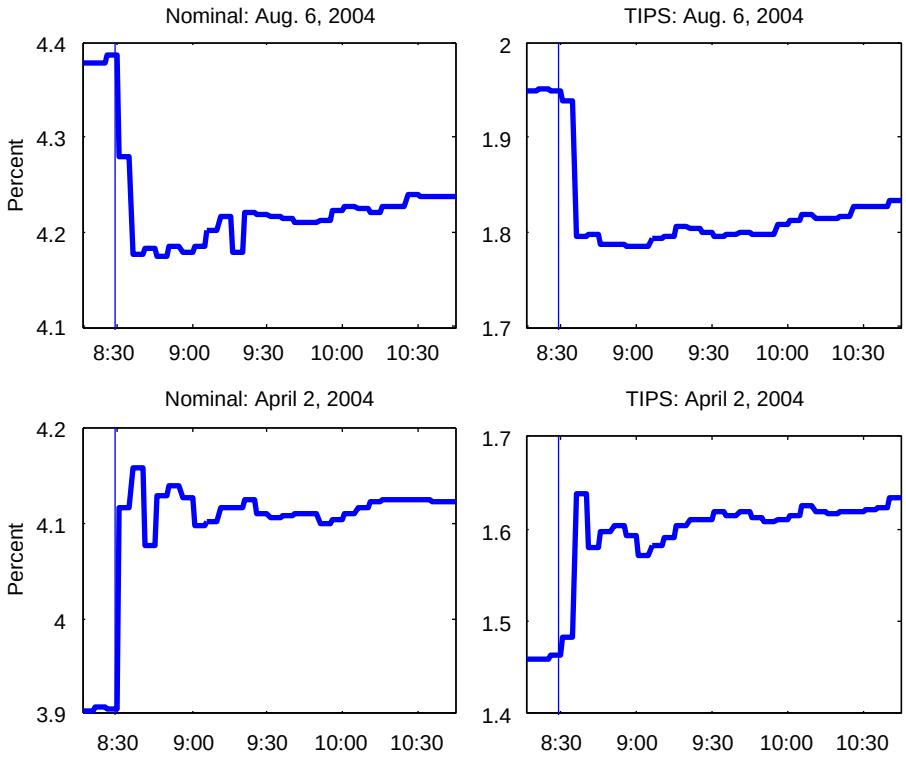


Fig. 1. Intradaily plots of ten-year nominal and TIPS yields around two employment reports. Notes: the employment reports released on April 2, 2004 and August 6, 2004 were the two largest absolute surprises among employment reports in the sample. The August 6, 2004 report was much weaker than expected. The April 2, 2004 report came in well above expectations.

can be used to construct five-to-ten-year forward nominal and real interest rates.[2] In addition, five-year, ten-year, and five-to-ten-year forward inflation compensation, defined as the spreads between nominal and comparable-maturity real yields can be computed.[3] On-the-run nominal securities command a liquidity premium, and so trade at lower yields than their off-the-run counterparts. However, in the narrow windows around news announcements, the relative liquidity of on- and off-the-run bonds is unlikely to change. Likewise, a liquidity premium tends to push up TIPS yields relative to their nominal counterparts; but again, this liquidity premium should difference out under the plausible assumption that macroeconomic news does not affect the relative liquidity of index-linked and nominal securities.

3. Regressions and results

As a preliminary illustration of how macroeconomic news affects nominal and index-linked bond yields, Fig. 1 plots the ten-year nominal and TIPS yields from 8:15 am to 9:30 am on the days of the two largest employment report surprises in our sample: April 2, 2004, when nonfarm payrolls growth was 208,000 above survey expectations, and August 6, 2004, when payrolls came in 193,000 below expectations.

As one might expect in an efficient market, yields jumped up on the stronger-than-expected data and down on the weaker-than-expected data, with the reactions complete within 10 min. Such rapid reaction of nominal interest rates is well documented in the news-announcement literature. But what has not been documented before is that TIPS yields move just as rapidly. The magnitude of the jump in TIPS yields on surprise days was similar to – but a bit smaller than – that in nominal yields. Having illustrated the point with these two large data surprises, we now turn to a more systematic analysis of the relationship between surprises and high-frequency yield changes.

[2] The approximation is that if y_n denotes the yield on a bond with a maturity of n years and duration D_n, then the five-to-ten-year forward rate is approximately $(D_{10}y_{10}-D_5y_5)/(D_{10}-D_5)$.

[3] All yields are expressed with semiannual compounding, and the calculation of the spread takes appropriate account of this. That is, if y_n^{nom} and y_n^{real} denote the nominal and real yield in percentage points, inflation compensation is constructed as $200\left(1+y_n^{nom}/200/1+y_n^{real}/200-1\right)$.

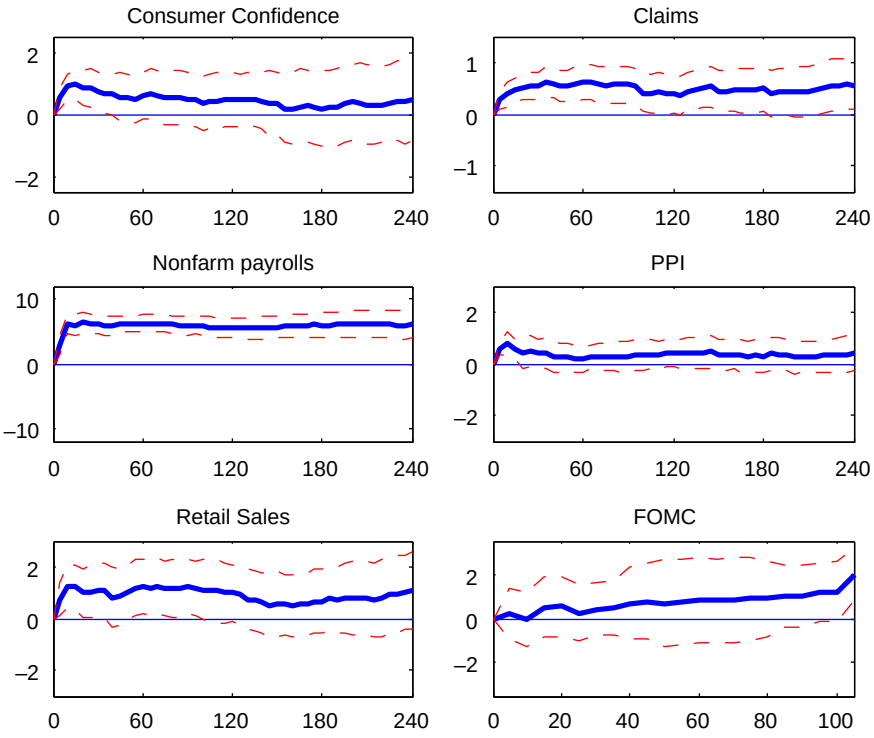


Fig. 2. Estimated effects of selected news surprises on nominal ten-year yields. Notes: the figure plots the estimated coefficients $\beta_j(h)$ in Eq. (1) against h, for six news announcements, where the dependent variable is the intradaily change in ten-year nominal yields. The window used in the regression is from 15 min before the announcement time until h min afterwards, where h is up to 4 hours (except for the FOMC announcements where it is up to 1 hour and 45 min after the 2:15 pm announcement).

3.1. Regressions of changes in yields and forward rates on surprises

The regressions considered in this paper are of the following form:

$$\Delta y_t(h) = \sum_{j=1}^{J} \beta_j(h) s_{j,t} + \varepsilon_t \tag{1}$$

where $\Delta y_t(h)$ is the change in some yield or forward rate from 15 min before a news announcement until h min afterwards, and the variable $s_{j,t}$ is the standardized surprise component of an announcement of type j. The regression is run over all windows where there is at least one news announcement: the surprise is set to zero for news types for which there is no announcement during that window. The device of regressing yield changes on surprise components of all different types of announcements jointly accommodates the possibility of multiple announcements coming out concurrently, as often happens (for example, the unemployment report and payrolls releases always come out at the same time).[4]

3.2. Documenting the jump behavior of nominal and TIPS yields

Figs. 2 and 3 plot the estimates of the coefficients $\beta_j(h)$ against h for some of the news announcements, for changes in ten-year nominal and TIPS yields, respectively. These can be thought of as plotting the expected effect of a one standard deviation surprise on yields at different times after the news announcement. It is evident from these figures that the impact of news announcements on nominal and TIPS yields is of the nature of a jump in the conditional mean, consistent with the example in Fig. 1. This result is well known for nominal yields. Fleming and Remolona (1999), using very high-frequency data on nominal Treasury securities, find that the adjustment in the conditional mean is complete within a couple of minutes. Andersen et al. (2007) and Ederington and Lee (1993) document similar jump behavior for other nominal interest rates. And it is what one would expect in an efficient market, following the reasoning of French and Roll (1986).

The reaction of ten-year TIPS yields is as fast as the reaction of nominal yields, judging by our five-minutely data. For TIPS yields, one might have worried that the small size and comparatively lower liquidity of this market would result in

[4] If no two types of news announcements ever came out concurrently, the OLS estimation of (1) could be performed simply as separate regressions of yield changes on surprises by type.

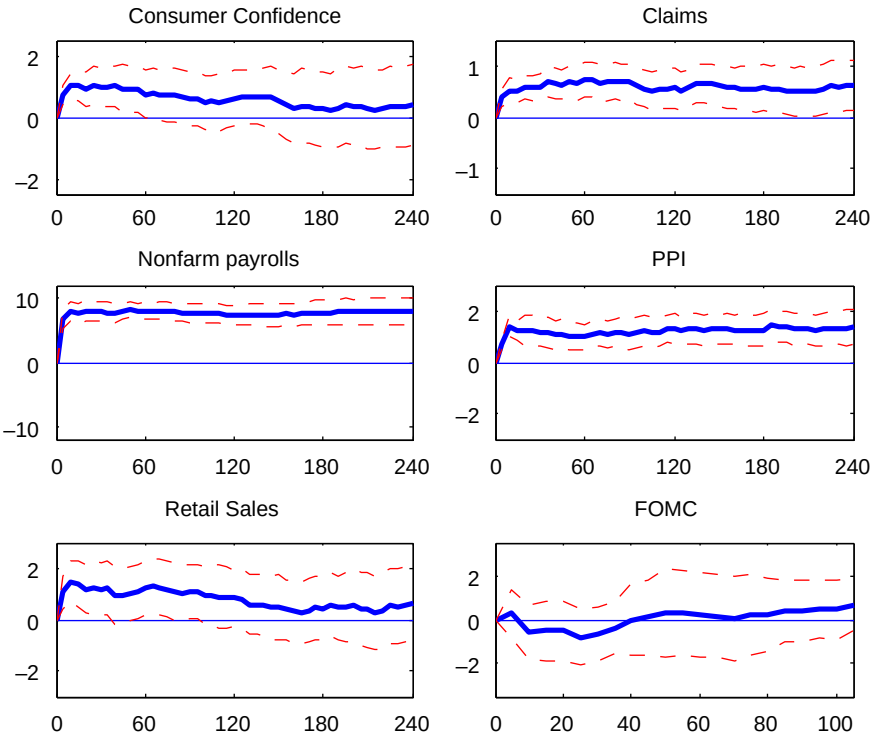


Fig. 3. Estimated effects of selected news surprises on ten-year TIPS yields. Notes: the figure plots the estimated coefficients $\beta_j(h)$ in Eq. (1) against h, for six news announcements, where the dependent variable is the intradaily change in ten-year TIPS yields. The window used in the regression is from 15 min before the announcement time until h min afterwards, where h is up to 4 hours (except for the FOMC announcements where it is up to 1 hour and 45 min after the 2:15 pm announcement).

a delayed reaction or some unwinding of initial jumps, but Figs. 1–3 show that this is not the case. In this context, it is worth pointing out that although the TIPS market is much smaller than the market for nominal Treasury securities, it cannot be thought of as a small or illiquid market. According to the FR-2004 survey of primary dealers conducted by the Federal Reserve Bank of New York, primary dealer trading volumes in TIPS averaged about $8 billion per day over the period covered by our intradaily dataset.

3.3. *Comparing the reaction of nominal and TIPS yields to news announcements*

This subsection compares the high-frequency reactions of nominal and TIPS yields of different maturities to all 14 news announcements. Table 3 reports coefficient estimates from estimating Eq. (1) with 9 different dependent variables: the changes in five-year and ten-year yields and the five-to-ten year forward rate on nominal bonds, TIPS, and inflation compensation, setting the window size h to 15 min. This is a conservative choice of window size, as it is safe to say that the jump in the conditional mean of nominal and TIPS yields is complete within 15 min of the data release.

Nearly all of the announcements cause statistically significant increases in five- and ten-year nominal yields (columns 1 and 4). Moreover, and as documented by others, the sensitivity is still evident in the five-to-ten-year forward nominal rates (column 7). Different patterns emerge, however, for different types of news announcements when nominal rates are decomposed into their real and inflation components.

3.3.1. *Price news*

Overall, it is inflation compensation rather than real rates that appear to be sensitive to news about prices. Core CPI announcements elicit statistically significant increases in inflation compensation at all maturities, including the far-forward rate (columns 3, 6 and 9 in Table 3), but do not draw a systematic reaction in TIPS yields. Higher-than-expected core PPI announcements lead to significant increases in five- and ten-year TIPS yields and inflation compensation, but at the five-to-ten-year horizon, the effect remains significant at the 5% level only for inflation compensation. These results are at once heartening and puzzling; heartening in that one would expect inflation surprises to have the most bearing on investors' inflation forecasts and their perception of inflation risk, and yet puzzling in that the reactions are still evident for forward rates five to ten years ahead, well after most inflation shocks should have died out.

Table 3
Regressions of intradaily yield changes on macroeconomic surprises.

	Ten-year			Five-year			Five-to-ten-year forward		
	Nominal (1)	TIPS (2)	IC (3)	Nominal (4)	TIPS (5)	IC (6)	Nominal (7)	TIPS (8)	IC (9)
Capacity	0.60***	0.39*	0.20	0.64***	0.36*	0.28	0.54***	0.46	0.07
	(0.19)	(0.23)	(0.16)	(0.23)	(0.20)	(0.19)	(0.18)	(0.36)	(0.26)
Confidence	1.02***	0.95***	0.06	1.03***	1.15***	−0.13	1.00***	0.72***	0.27*
	(0.21)	(0.19)	(0.05)	(0.24)	(0.25)	(0.11)	(0.23)	(0.20)	(0.15)
Core CPI	2.21***	0.64	1.55***	2.55***	0.79	1.74***	1.77***	0.40	1.35***
	(0.34)	(0.41)	(0.31)	(0.45)	(0.51)	(0.39)	(0.26)	(0.40)	(0.39)
Durable	0.93***	0.83***	0.09	1.03**	1.00***	0.02	0.79***	0.70**	0.08
	(0.35)	(0.28)	(0.11)	(0.41)	(0.30)	(0.22)	(0.31)	(0.35)	(0.22)
ECI	0.11	0.15	−0.04	0.10	0.37	−0.27	0.11	−0.01	0.11
	(0.23)	(0.17)	(0.14)	(0.36)	(0.22)	(0.26)	(0.21)	(0.23)	(0.21)
GDP	1.40***	0.97***	0.42*	1.73**	1.30***	0.42	0.94***	0.62*	0.31
	(0.45)	(0.32)	(0.23)	(0.80)	(0.42)	(0.48)	(0.25)	(0.34)	(0.25)
Claims	0.52***	0.46***	0.05	0.67***	0.49***	0.17**	0.32***	0.45***	−0.13
	(0.09)	(0.09)	(0.06)	(0.12)	(0.11)	(0.07)	(0.09)	(0.10)	(0.08)
NAPM	1.84***	1.67***	0.15**	2.15***	1.89***	0.24**	1.43***	1.46***	−0.05
	(0.26)	(0.27)	(0.07)	(0.41)	(0.42)	(0.10)	(0.21)	(0.19)	(0.17)
Nonfarm	7.50***	5.98***	1.44***	10.34***	7.37***	2.87***	3.76***	4.54***	−0.83
	(0.71)	(0.55)	(0.25)	(1.02)	(0.80)	(0.70)	(0.56)	(0.65)	(0.53)
New homes	0.66***	0.55***	0.10**	0.84***	0.63***	0.21**	0.43**	0.47***	−0.05
	(0.17)	(0.15)	(0.05)	(0.20)	(0.18)	(0.09)	(0.17)	(0.15)	(0.09)
Core PPI	1.25***	0.58***	0.66***	1.41***	0.66***	0.74***	1.06***	0.50*	0.55**
	(0.24)	(0.21)	(0.18)	(0.27)	(0.20)	(0.15)	(0.21)	(0.27)	(0.23)
Retail sales	1.40***	1.20***	0.18	1.58***	1.42***	0.14	1.17**	0.95**	0.20
	(0.54)	(0.41)	(0.16)	(0.58)	(0.46)	(0.17)	(0.52)	(0.42)	(0.22)
Unemployment	2.52***	2.17***	0.32**	3.57***	2.35***	1.19	1.15**	2.08***	−0.95
	(0.54)	(0.42)	(0.16)	(0.73)	(0.77)	(0.79)	(0.45)	(0.78)	(0.68)
FOMC	−0.49	0.49**	−0.98***	0.92	1.62**	−0.72***	−2.32***	−0.64	−1.65***
	(0.33)	(0.23)	(0.21)	(0.58)	(0.69)	(0.18)	(0.52)	(0.36)	(0.37)

Notes: Estimates of the coefficients in Eq. (1) using data from the intradaily database on five- and ten-year nominal and TIPS yields from 15 min before each announcement to 15 min after. White standard errors are shown in parentheses. Significance at the 10%, 5% and 1% levels are denoted by *, ** and ***, respectively. Sample period: February 17, 2004 to June 13, 2008.

3.3.2. *Real-side news*

Turning to news announcements concerning real economic activity, the decomposition of the response into real and inflation components is very different. Positive surprises to consumer confidence, durable goods orders, GDP, new home sales, initial claims, nonfarm payrolls, the NAPM index, retail sales and the unemployment rate all raise five- and ten-year TIPS yields and the coefficients are statistically significant at the 1% significance level (columns 2 and 5 in Table 3). Some of these announcements also have statistically significant effects on inflation compensation, but the reactions are modest: news about real activity seems to affect TIPS yields considerably more than inflation compensation. For example, a nonfarm payrolls release that is one standard deviation above expectations is estimated to increase ten-year TIPS yields by 6 basis points, while ten-year inflation compensation increases only 1.4 basis points, with both coefficients precisely estimated.

Forward rates are analytically useful because they abstract from the effects of variation in short-term interest rates and the business cycle on longer term yields and give a better view of investors' beliefs about the longer term. Specifically, distant-horizon real forward rates are comprised of long-run expectations of real short-term interest rates and real risk premia. Meanwhile, forward rates of inflation compensation comprise long-run inflation expectations and inflation risk premia. The results in Table 3 (columns 7, 8 and 9) indicate that the sensitivity of nominal forward rates to news about the real side of the economy is heavily concentrated in real forward rates. Of the 11 indicators of real economic activity in our study, 6 have significant positive effects on five-to-ten-year forward TIPS yields at the 1% significance level, and 2 more are significant at the 5% level. None have a significant effect on five-to-ten-year forward inflation compensation, even at the 5% level.

3.3.3. *Monetary policy news*

Monetary-policy surprises have no statistically significant effect on five- or ten-year nominal yields in Table 3. But this masks offsetting effects on real and inflation components. As can be seen in this Table, a surprise tightening of policy leads to an increase in five-year real yields, a smaller rise in ten-year real yields, and a decline in both five- and ten-year inflation compensation. All of these effects are statistically significant. Further ahead, at the five-to-ten-year horizon, the point estimate of the TIPS response becomes negative but is not significant, while the decline in inflation compensation becomes more pronounced. Given that the standard deviation of a funds rate surprise is 5.3 basis points (Table 1), the interpretation of our coefficient estimates is that a 25 basis point surprise tightening rotates the nominal yield curve, lowering five-to-ten-year forward nominal rates by 12 basis points, of which 3 and 9 basis points are real and inflation compensation,

respectively. A natural interpretation is that a surprise policy tightening leads investors to mark up their expectations for the real path of policy in the near-term, and perhaps also leads to some increase in real risk premia. However, the surprise tightening leads investors to revise down their inflation expectations, in both the short- and long-run, and reduce the compensation demanded for inflation risk, leaving inflation compensation lower at all horizons.

The reaction to monetary-policy surprises is worth commenting upon a little more. It is central to the standard view of the transmission mechanism of monetary policy that tightening actions by the central bank should raise benchmark long-term interest rates, and hence raise the borrowing costs for households and businesses that are tied to those benchmark rates. Empirically, this relationship has been hard to detect. Some authors, including Cook and Hahn (1989) and Roley and Sellon (1995), have found little evidence of a relationship between Federal Reserve policy actions and longer term nominal Treasury yields. Yet we find that unexpectedly tighter monetary-policy raises *real* rates, which are presumably what matter for the spending decisions of agents, providing support for the standard view of the transmission mechanism of monetary policy. Because unexpectedly tighter policy also appears to lower inflation expectations and/or inflation risk premia, the increase in real rates is obscured, leaving no statistically significant effect on long-term nominal interest rates in our sample.[5] It remains an open question as to why far-forward rates of inflation compensation respond to today's monetary-policy surprises, a question discussed further in Section 4.

3.4. Comparing intradaily and daily results

Gürkaynak et al. (2005a) and Gürkaynak et al. (2009b) have performed similar analysis of the response of real yields and inflation compensation to macro announcement surprises, but have done so using daily-frequency data. However, as argued by Andersen et al. (2003), Gürkaynak et al. (2005b) and others, high-frequency intradaily data, as used in this paper, have the potential to give more precise estimates of announcement effects than can be obtained with daily data.

To measure the gains that can be obtained from high-frequency data on TIPS yields, regression (1) is adapted to use daily yield changes as the dependent variable in place of changes in narrow intradaily windows. Table 4 shows the results using the 4 pm observation each day on the specific five- and ten-year nominal and TIPS bonds in our intradaily dataset. Not surprisingly, the standard errors are much larger, and several of the announcement effects are no longer statistically significant. Averaging across all the standard errors in Table 3 (the intradaily data) gives a standard error of 0.31. The corresponding average standard error in Table 4 (the daily data) is 0.73—more than twice as big. The results with the intradaily and daily datasets are not inconsistent; it is simply that the latter are much less precise. For example, using the intradaily data, the 95% confidence interval for the effect of a one standard deviation retail sales surprise on ten-year TIPS yields is from 0.4 to 2.0 basis points. The corresponding confidence interval using daily data is from 0.7 to 4.1 basis points. As a result of this loss of precision, the number of announcements that have a significant effect on ten-year TIPS yields at the 1% significance level drops from 10 using the intradaily data to 5 with the daily data.

A more formal test of the hypothesis that intradaily and daily data are estimating the same parameter, but that the intradaily estimate is more precise, is to apply Hausman specification tests to the two coefficient estimates. Given two estimators, one less efficient than the other, the null of the Hausman test is that both estimators are consistent for the same parameter. In our context, the null amounts to the hypothesis that the mean reaction to news surprises is the same in short windows as it is for whole days, implying that nominal and TIPS yields exhibit jump behavior in response to news announcements. The coefficient of the intradaily regression is assumed to be the more efficient estimator. The Hausman test statistic is calculated for each pair of coefficients, i.e., for a given maturity interest rate and a specific news variable, resulting in 126 test statistics.[6] Of these, only 14 reject the null hypothesis at the 5% level and 7 at the 1% level. The vast majority cannot reject the null, leaving us confident that the intradaily regressions simply provide more efficient estimates of the reactions to news announcements than daily data.[7]

4. Interpretation and conclusions

The high-frequency reaction of asset prices to macroeconomic news is a rich source of information about the effects of economic fundamentals on financial markets; reactions that are clouded with lower frequency data come into sharper focus with intradaily data. With intradaily quotes on nominal government yields, we confirm the finding that long-term nominal yields and forward rates are very sensitive to macroeconomic news announcements. As an accounting matter,

[5] Our results differ from Kuttner (2001) who, using an earlier sample of daily data, found that unexpected monetary policy tightenings raised long-term nominal yields.

[6] The general form of the Hausman specification test can be found in Hausman (1978). For our application, the test statistic has a χ^2 null distribution with one degree of freedom. In four cases the test statistic is not defined because the estimated variance in the denominator of the test statistic is negative.

[7] The appendix available online also gives results for estimating Eq. (1) with daily changes in nominal and TIPS yields from the smoothed fitted yield curves estimated late each afternoon by Federal Reserve staff. These curves are described by Gürkaynak et al. (2007, 2009b). The results are very similar to the estimates reported in Table 4. Appendix A also documents evidence of a structural break in the relationship between TIPS yields and news announcements before 2004, perhaps the consequence of the TIPS market having become larger and more mature.

Table 4
Regressions of daily yield changes on macroeconomic surprises.

	Ten-year			Five-year			Five-to-ten-year forward		
	Nominal	TIPS	IC	Nominal	TIPS	IC	Nominal	TIPS	IC
Capacity	0.46	0.26	0.20	0.63	0.53	0.09	0.39	−0.26	0.64**
	(0.55)	(0.46)	(0.38)	(0.67)	(0.55)	(0.55)	(0.44)	(0.42)	(0.31)
Confidence	0.83	0.33	0.49	0.30	0.78	−0.48	1.48	−0.27	1.73
	(0.81)	(0.78)	(0.51)	(0.93)	(0.99)	(0.60)	(1.06)	(0.88)	(1.39)
Core CPI	1.57**	−0.48	2.04***	1.96**	−1.01	2.96***	1.31*	0.15	1.15***
	(0.78)	(0.58)	(0.45)	(0.90)	(0.71)	(0.55)	(0.75)	(0.55)	(0.41)
Durable	1.22	1.33*	−0.13	1.73*	2.37***	−0.67	0.56	0.67	−0.12
	(0.83)	(0.71)	(0.41)	(1.00)	(0.89)	(0.60)	(0.68)	(0.69)	(0.43)
ECI	0.07	2.51**	−2.45**	−0.18	2.66*	−2.86***	0.30	2.51*	−2.21**
	(1.88)	(1.26)	(0.98)	(2.08)	(1.41)	(0.93)	(1.72)	(1.39)	(1.08)
GDP	2.21	2.26***	−0.07	2.51	2.57***	−0.10	1.28	1.95**	−0.69
	(1.44)	(0.85)	(1.00)	(1.76)	(0.99)	(1.11)	(1.27)	(0.87)	(0.85)
Claims	1.21***	0.89**	0.31*	1.24***	0.90**	0.33	1.23***	0.91**	0.31
	(0.41)	(0.36)	(0.17)	(0.45)	(0.40)	(0.22)	(0.41)	(0.36)	(0.24)
NAPM	2.16***	2.45***	−0.32	2.47***	2.90***	−0.46	1.75***	2.00***	−0.27
	(0.63)	(0.51)	(0.38)	(0.77)	(0.75)	(0.47)	(0.65)	(0.51)	(0.40)
Nonfarm	8.19***	6.27***	1.83***	10.06***	7.78***	2.18***	5.77***	4.72***	0.97**
	(1.11)	(0.94)	(0.40)	(1.24)	(1.07)	(0.51)	(1.01)	(0.93)	(0.38)
New homes	0.82*	0.50	0.32	1.04*	0.78*	0.24	0.53	0.54	−0.02
	(0.48)	(0.39)	(0.28)	(0.54)	(0.46)	(0.36)	(0.46)	(0.50)	(0.37)
Core PPI	1.76***	0.57	1.18***	1.87**	0.48	1.37***	1.68***	0.76**	0.91**
	(0.66)	(0.37)	(0.45)	(0.76)	(0.47)	(0.45)	(0.58)	(0.36)	(0.44)
Retail sales	2.20**	2.42***	−0.24	2.64**	2.51***	0.10	1.66	2.25**	−0.61
	(1.05)	(0.86)	(0.43)	(1.20)	(0.93)	(0.59)	(1.08)	(0.94)	(0.41)
Unemployment	3.99***	2.96***	0.99*	5.10***	3.90***	1.14	2.51***	2.05***	0.43
	(0.71)	(0.78)	(0.56)	(0.77)	(0.87)	(0.71)	(0.77)	(0.78)	(0.40)
FOMC	0.14	0.41	−0.27	1.58	2.20**	−0.64	−2.37***	−1.10*	−1.24
	(0.91)	(0.56)	(0.82)	(1.24)	(0.87)	(0.54)	(0.79)	(0.66)	(0.90)

End-of-day data on securities from the intradaily database.
Notes: Estimates of the coefficients in Eq. (1) using data from the intradaily database on five- and ten-year nominal and TIPS yields from 4pm on the day before each announcement to 4pm on the announcement day. White standard errors are shown in parentheses. Significance at the 10%, 5% and 1% levels are denoted by *, ** and ***, respectively. Sample period: February 17, 2004 to June 13, 2008.

such sensitivity must reside in some combination of expectations of future real short-term interest rates or inflation, or in real or inflation risk premia. Our new dataset of intradaily quotes on real indexed government debt allows us to parse the reactions into real and inflation components more precisely than has been possible until now.

News about prices, the real side of the economy, and monetary policy elicit distinctly different responses from real rates and rates of inflation compensation. Price data mainly affect inflation compensation, indicating that inflation expectations are poorly anchored, or that the premium required to hold inflation risk over the long run is sensitive to today's inflation developments. Tighter than expected monetary policy lowers far-forward inflation compensation, adding to that picture. Were agents' perceptions of the Fed's implicit inflation target time varying, as in the model of Gürkaynak et al. (2005a), monetary-policy surprises could prompt revisions to long-run inflation expectations. Alternatively, inflation expectations may be reasonably stable, but inflation uncertainty and hence risk premia may be sensitive to unexpectedly higher inflation today or unexpectedly looser monetary policy.

Real-side data have a large impact on real yields and forward rates—but have no measurable effect on forward rates of inflation compensation. Moreover, the per-quarter standard deviation of real and nominal forward rates due to *all* 14 announcements collectively (including real activity, price and monetary-policy announcements) can be computed—the standard deviation of real forward rates is about three times larger than that of forward rates of inflation compensation. In this sense the overall sensitivity of forward rates to news announcements is more on the real side than in inflation compensation. Evidently, expectations of short-term real interest rates far into the future and/or long-term real risk premia are substantially buffeted by incoming data. Explanations for the sensitivity of nominal forward rates to news that hinge upon unanchored long-run inflation expectations alone are insufficient.

The sensitivity of real rates to incoming economic data is hard to reconcile within the framework of most theoretical models as they predict that real interest rates are stationary, and thus that arbitrarily distant-horizon expectations of real short-term interest rates ought to be constant. Our finding is, in particular, at odds with widely used dynamic stochastic general equilibrium models, including those of Smets and Wouters (2007) and Christiano et al. (2005). Such models imply that the long-run expected value of the real short-term interest rate is constant and that even near-term expectations of real short-term interest rates are very stable and so should react little to incoming data. In a Ramsey model framework – in which equilibrium real rates depend on the trend growth rate of potential output, population growth and agents'

preferences – it is hard to see how one month's data could have more than a tiny effect on the ten-year-ahead expected real short-term rate of interest.

In principle, it is possible that long-term expectations of real short-term interest rates are sensitive to news announcements because of the role of learning about the trend growth rate of potential output (e.g., Edge et al. 2007), or because there is a small permanent component of consumption growth and investors have preferences for the early resolution of uncertainty (as in the models of Bansal and Yaron, 2004; Bansal and Shaliastovich, 2008). However, long-term real forward rates routinely jump 10 basis points on a single employment report, and this seems hard to explain within the framework of any of these models, unless perhaps there is some constant-gain learning in which agents forget the past very quickly.

Beyond expectations of real short-term interest rates, the sensitivity of real forward rates to news announcements may also owe to time-variation in real risk premia. This paper finds that most of the sensitivity of nominal rates to real-side surprises is in real rates, while Beechey (2007) found that most of it was in term premia. If both results are correct, then most of the puzzling sensitivity to these news announcements must be in real term premia. This would still be puzzling, as real risk premia are generally thought of as being countercyclical, and so should, if anything, be pushed *down* by news indicating stronger-than-expected economic activity.

It is easy to argue why any one of the components of distant-horizon nominal forward rates should *not* react much to individual news announcements—but these forward rates do react to news, and the reaction has to be in one or more of the components. What can be concluded is that the sizeable impact of news about real activity on the nominal term structure of interest rates is concentrated in the real component. This suggests that long-run inflation expectations are better anchored than one would infer from studying the effects of news on distant-horizon nominal forward rates alone, and assuming that this reflected only variation in inflation expectations. It also creates a puzzle for macroeconomic models in that distant-horizon real rates are sensitive to data releases to an extent that seems hard to reconcile within the framework of existing macroeconomic models.

Appendix A. Supplementary data

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jmoneco.2009.03.011.

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